창의공학 설계
모터 실습



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실험 목표

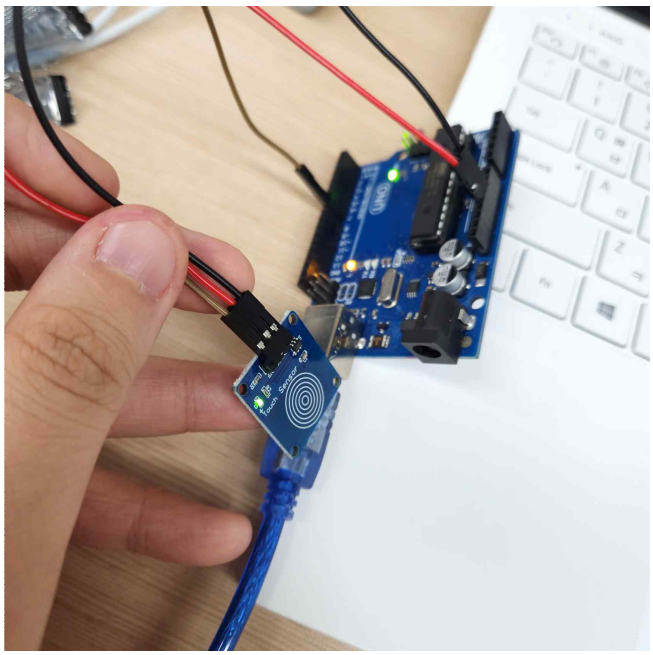
- ✓ 기초 실습 1 - 터치 센서
- ✓ 기초 평가 1 - 터치 센서 & LED 제어
- ✓ 기초 실습 2 - 부저 센서 & 조이스틱 센서
- ✓ 기초 평가 2 - 조이스틱 & LED 제어
- ✓ 응용 평가 - MP3 만들기

실험 과정

기초 실습 1. 터치 센서

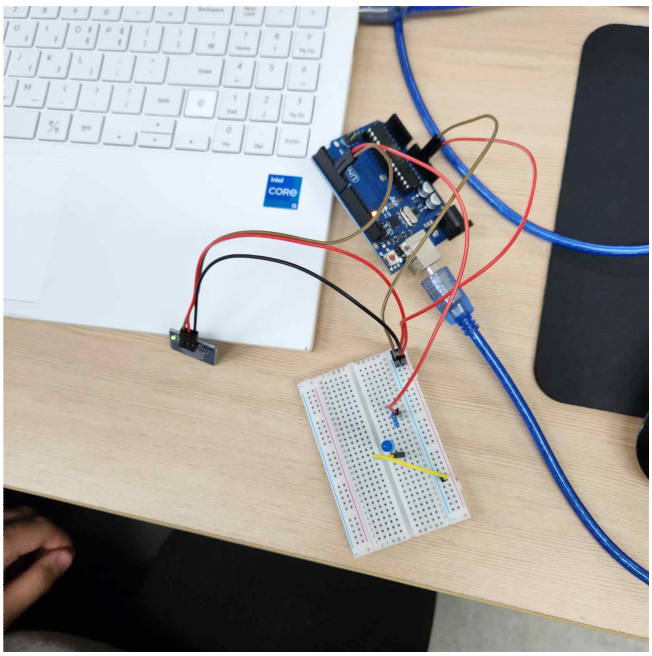
코드	회로도

```
int touchPin = 7;
int state = 0;
bool touchFlag = false;
void setup()
{
  Serial.begin(9600);
  pinMode(touchPin, INPUT);
}
void loop()
{
  state = digitalRead(touchPin);
  if (state == HIGH)
  {
    if (touchFlag == false)
    {
      touchFlag = true;
      Serial.println("on");
    }
  }
  else
  {
    if (touchFlag == true)
    {
      touchFlag = false;
      Serial.println("off");
    }
  }
}
```



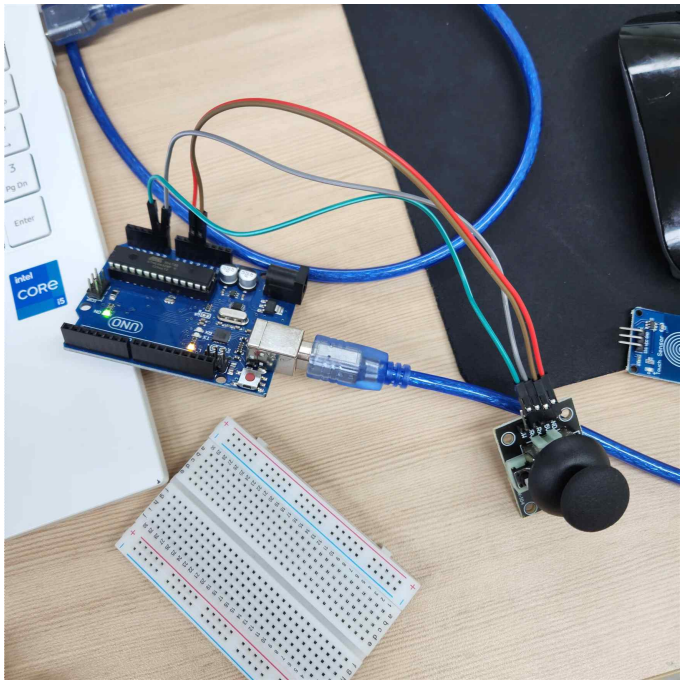
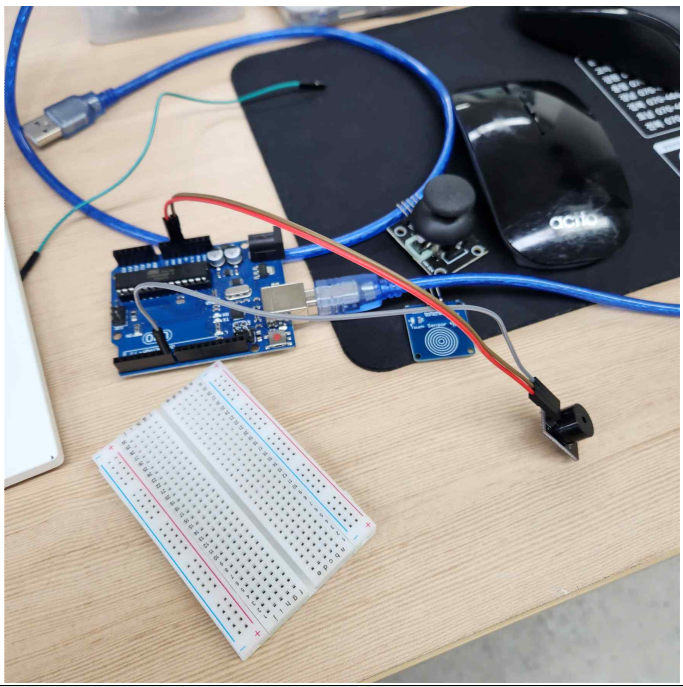
기초 평가 1. 터치 센서 & LED 제어

코드	회로도

int touchPin = 5;
int ledPin = 4;
int state = 0;
bool touchFlag = false;
void setup()
{
  Serial.begin(9600);
  pinMode(touchPin, INPUT);
  pinMode(ledPin, OUTPUT);
}
void loop()
{
  state = digitalRead(touchPin);
  if (state == HIGH)
  {
  if (touchFlag == false)
  {
  touchFlag = true;
  digitalWrite(ledPin, HIGH);
  }
  }
  else
  {
  if (touchFlag == true)
  {
  touchFlag = false;
  digitalWrite(ledPin, LOW);
  }
  }
}
	

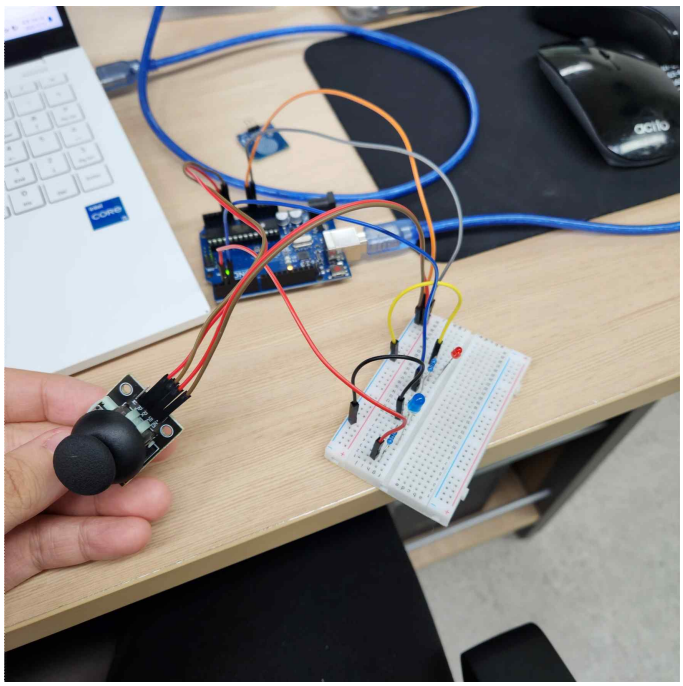
기초 실습 2. 부저 센서, 조이스틱 센서

코드	회로도

# 조이스틱 센서 	
`int x;`
`int y;`
`void setup()`
`{`
`  Serial.begin(9600);`
`  pinMode(8, INPUT_PULLUP);`
`}`
`void loop()`
`{`
`  x = analogRead(A0);`
`  y = analogRead(A1);`
`  Serial.print(x);`
`  Serial.print("  ");`
`  Serial.print(y);`
`  Serial.print("  ");`
`  Serial.println(digitalRead(8));`
`  delay(100);`
`}`
 	
# 부저 센서 	
`int pin_num = 7;`
`void setup()`
`{`
`  Serial.begin(9600);`
`}`
`void loop()`
`{`
`  tone(pin_num,1000,500);`
`  delay(500);`
`  tone(pin_num,300,500);`
`  delay(500);`
`}`
	#조이스틱 센서
	 
	
	#부저 센서
	 

기초 평가 2. 조이스틱 & LED 제어

코드	회로도

int x;
int y;
int redLED = 3;
int blueLED = 2;
void setup()
{
  Serial.begin(9600);
  pinMode(8, INPUT_PULLUP);
  pinMode(redLED, OUTPUT);
  pinMode(blueLED, OUTPUT);
}
void loop()
{
  x = analogRead(A0);
  y = analogRead(A1);
  Serial.print(x);
  Serial.print(" ");
  Serial.print(y);
  Serial.print(" ");
  Serial.println(digitalRead(8));
  if (x == 1023)
  {
  digitalWrite(blueLED, HIGH);
  digitalWrite(redLED, LOW);
  }
  else if (x == 0)
  {
  digitalWrite(redLED, HIGH);
  digitalWrite(blueLED, LOW);
  }
  else
  {
  digitalWrite(redLED, LOW);
  digitalWrite(blueLED, LOW);
  }
  delay(100);
}
	

응용 평가. MP3 만들기

코드

```
#define JOYSTICK_X A0
#define JOYSTICK_Y A1
#define BUZZER_PIN 8

int length_star = 15;
char notes_star[] = "ccggaag ffeeddc ggffeed ggffeed ccggaag ffeeddc ";
int beats_star[] = { 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 2, 4 };
int tempo_star = 300;

int length_jingle = 26;
char notes_jingle[] = "eeeeeeegcde fffffeeeeddedg";
int beats_jingle[] = { 1, 1, 2, 1, 1, 2, 1, 1, 1, 1, 4, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 2, 2};
int tempo_jingle = 300;

int selectedSong = 0;
bool isPlaying = false;

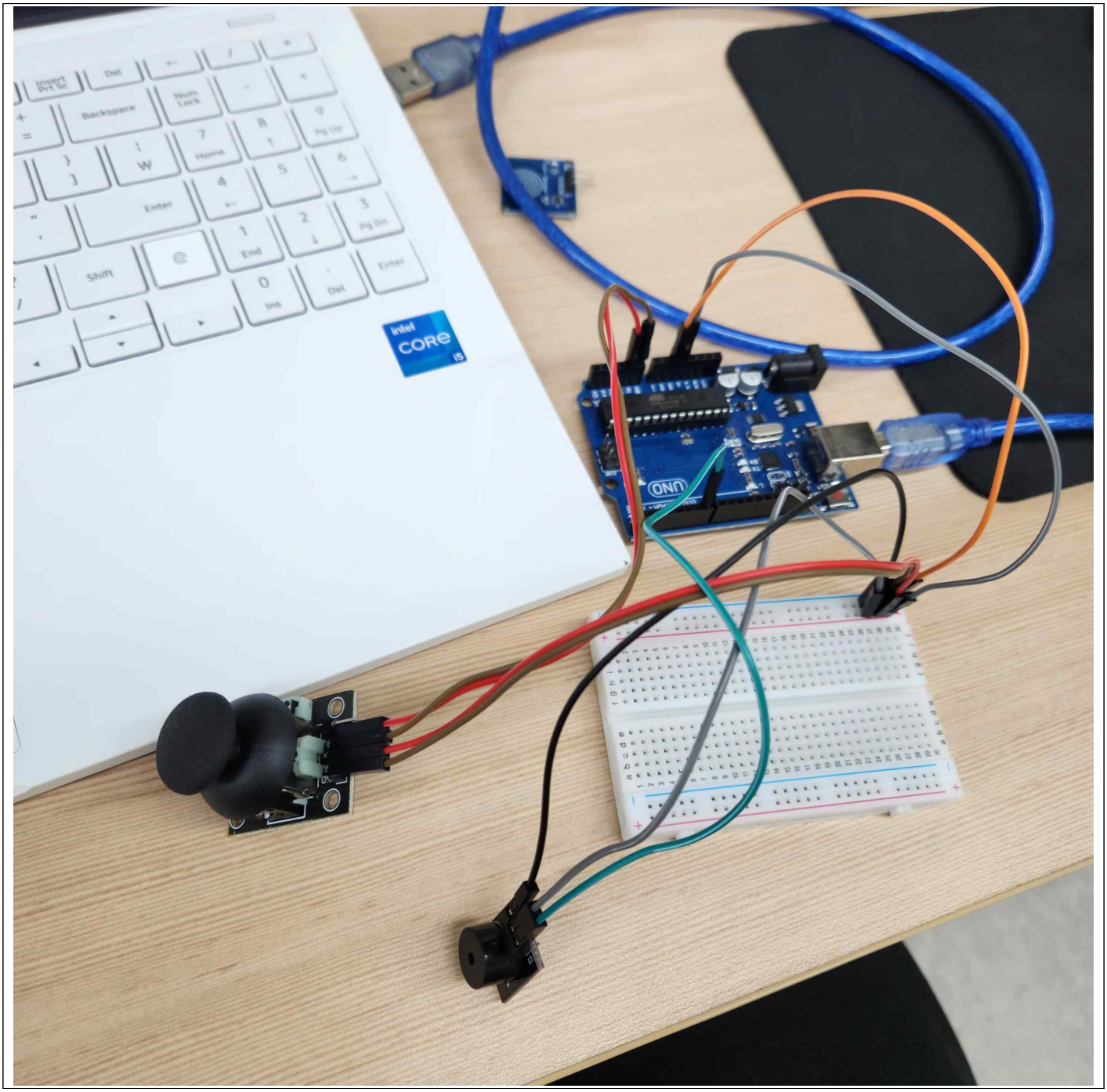
void playTone(int tone, int duration)
{
  for (long i = 0; i < duration * 1000L; i += tone * 2)
  {
    digitalWrite(BUZZER_PIN, HIGH);
    delayMicroseconds(tone);
    digitalWrite(BUZZER_PIN, LOW);
    delayMicroseconds(tone);
  }
}
void playNote(char note, int duration)
{
  char names[] = { 'c', 'd', 'e', 'f', 'g', 'a', 'b', 'C' };
  int tones[] = { 1915, 1700, 1519, 1432, 1275, 1136, 1014, 956 };
  for (int i = 0; i < 8; i++)
  {
    if (names[i] == note)
    {
      playTone(tones[i], duration);
    }
  }
}
void playSelectedSong()
```

```
{
  Serial.print("선택된 노래: ");
  switch (selectedSong)
  {
    case 1:
      Serial.println("반짝반짝 작은 별");
      for (int i = 0; i < length_star; i++)
      {
        if (notes_star[i] == ' ')
        {
          delay(beats_star[i] * tempo_star);
        }
        else
        {
          playNote(notes_star[i], beats_star[i] * tempo_star);
        }
        delay(tempo_star / 2);
      }
      break;
    case 2:
      Serial.println("징글벨");
      for (int i = 0; i < length_jingle; i++)
      {
        if (notes_jingle[i] == ' ')
        {
          delay(beats_jingle[i] * tempo_jingle);
        }
        else
        {
          playNote(notes_jingle[i], beats_jingle[i] * tempo_jingle);
        }
        delay(tempo_jingle / 2);
      }
      break;
    default:
      noTone(BUZZER_PIN);
  }
}
void setup()
{
  Serial.begin(9600);
  pinMode(JOYSTICK_X, INPUT);
  pinMode(JOYSTICK_Y, INPUT);
```

```
  pinMode(BUZZER_PIN, OUTPUT);
}
void loop()
{
  int xValue = analogRead(JOYSTICK_X);
  int yValue = analogRead(JOYSTICK_Y);
  static bool songChanged = false;
  if (!isPlaying)
  {
    int mappedSong = map(xValue, 0, 1023, 1, 2);
    if (selectedSong != mappedSong && !songChanged)
    {
      selectedSong = mappedSong;
      songChanged = true;
      switch (selectedSong)
      {
        case 1:
          Serial.println("반짝반짝 작은 별");
          break;
        case 2:
          Serial.println("징글벨");
          break;
        default:
          Serial.println("노래 없음");
      }
    }
    if (yValue < 200)
    {
      delay(500);
      playSelectedSong();

      songChanged = false;
    }
  }
}
```

회로도



-**고찰** : x축으로 노래의 종류를 바꾸고, y축을 작동시켜 노래가 재생되게 하였다.